

Behrad Rabiei

📍 Ann Arbor ✉ brabiei@umich.edu in behrad-rabiei 🌐 behradrabiei 📄 Google Scholar

Education

University of Michigan - Ann Arbor

Doctor of Philosophy (Ph.D.) - Robotics

Aug 2025 – June 2030

(Expected)

- GPA: N.A./4.0
- Advisors: Dr. Maani Ghaffari (Primary) and Dr. Yulun Tian
- Lab: Computational Autonomy and Robotics Laboratory (CURLY)

University of California - San Diego

MS in Electrical Engineering - Intelligent Systems, Robotics, and Controls

Sept 2023 – June 2024

- GPA: 3.744/4.0
- Lab: Existential Robotics Laboratory (ERL)

University of California - San Diego

BS in Electrical Engineering - Machine Learning and Controls

June 2020 – June 2023

- GPA: 3.618/4.0

Technologies

Hardware: Intel RealSense, 3D LiDARs, VectorNav, Nvidia Jetson, Raspberry Pi

Robots: F1 Tenth, Jackal UGV, Turtlebot4, DJI Custom Drones, Amazon Differential Drive Robots

Software: Python, C++, C, ROS, Open3D, PyTorch, JAX, GTSAM, Spot, LLMs (LangChain, LangFlow)

Tools: Git, Linux, Docker, Kubernetes, Gazebo

Research Interests

- Simultaneous Localization and Mapping (SLAM), Novel Environment Representations, Sensor Fusion for SLAM, LLM equipped Planning and Navigation
- Autonomous Robot Exploration, Perception-aware Planning and Control, Active SLAM, Model-based and Model-free Active Target Tracking
- Multi-robot Systems, Distributed Estimation and Planning
- **Relevant Fields:** Robotics, Machine Learning, Computer Vision, Distributed Optimization, Automation

Experience

Solutions Engineer

ARM

San Diego, CA

January 2025 – Sept 2025

- Co-manage a robust TCL codebase to automate power grid insertion, streamlining chip design workflows
- Contributing to the architecture and implementation of the second-generation power grid insertion system and hierarchical floorplanning methodology
- Leading a machine learning initiative to proactively identify wiring routes prone to timing violations, aiming to enhance design reliability and reduce verification time

Graduate Researcher

Existential Robotics Laboratory (UCSD)

San Diego, CA

July 2024 – Sept 2025

- Implemented a custom A* path planning algorithm with heuristic for LTL-based planning using semantically annotated 2D maps
- Implemented ROS1 nodes for A*, trajectory tracking, and differential drive control on Jackal UGV and TurtleBot 4, tested and validated the trajectories in Gazebo
- Built a real-time semantically annotated OctoMap from depth and semantic segmentation data using the

Semantic Shannon Mutual Information (SSMI) Library and YOLO

- Used Spot to generate automata from LTL formulas based on natural language queries

Software Engineer

PMAT

San Diego, CA

April 2024 – Jan 2025

- Debugged issues in software for Navy ships written in C++ and Java in a 40-year-old codebase
- Rebuilt and validated code, ensuring fixes meet performance and operational standards
- Wrote test procedures for engineers to verify software fixes
- Utilized version control to manage changes across a large, legacy codebase

Machine Learning Intern

NASA

Manhattan, NY

June 2024 – August 2024

- Developed a data retrieval system for CMIP 6, managing 30PB of climate data
- Created an efficient parser to filter datasets from over 13.6 million entries
- Integrated LangChain and OpenAI APIs to build a dynamic user-driven analysis engine
- Designed a Streamlit UI for a conversational, intuitive data analysis experience
- Submitted the project to the AGU 2024 and NeurIPS 2024 conferences for presentation

Machine Learning Intern

RoboCoach

San Diego, CA

May 2023 – May 2024

- Designed an open-source project that generates Python code from natural language
- Integrated LangChain and OpenAI APIs to enable code generation via multiple LLMs
- Developed custom prompts and used regex to extract code from plain text
- Created a chatbot-like web interface using Streamlit for a seamless user experience

System Architecture Co-Op

Amazon Robotics

North Reading, MA

July 2022 – Sept 2022

- Designed test plans to match exit criteria for Amazon's autonomous robot, Proteus
- Initiated cross-team collaboration to validate metrics and automate logging
- More than doubled facility throughput by expanding test scenarios and maps (from 6 to 15)
- Wrote Python scripts to extract real-time test data from AWS databases using SQL queries

Reliability Engineer Intern

Takeda

Glendale, CA

June 2021 – Sept 2021

- Liaised between Takeda and the vendor to shift from preventive to predictive maintenance
- Coordinated sensor installations, balancing vendor and production needs
- Secured approvals for shutdowns to enable smooth sensor deployment
- Helped draft a proposal that secured funding for expanded sensor installations
- Highlighted operational redundancies, showing continued production capability during installations and potential equipment failures
- Performed cost analysis, outlining the benefits of predictive maintenance adoption

Manufacturing Intern

SpaceX

Hawthorne, CA

July 2020 – Sept 2020

- Optimized the solenoid valve manufacturing process to boost production and quality
- Increased production rate by 114%, from 70 to 150 valves/day
- Added two new assembly steps to reduce assembly time
- Identified root causes of valve failures, addressing around 60 daily failures
- Created a corrective action guide for technicians to resolve production losses

Publications

LTLCCodeGen: Code Generation of Syntactically Correct Temporal Logic for Robot Task Planning

[PDF](#) [↗](#) *Spring 2025*

*Behrad Rabiei**, Mahesh Kumar A.R. *, Zhirui Dai, Surya L.S.R. Pilla, Qiyue Dong, Nikolay Atanasov

International Conference on Intelligent Robots and Systems (IROS) 2025

Climate PAL: Climate Analysis through Conversational AI

[PDF](#) [↗](#) *Fall 2024*

Sonia Crompt, *Behrad Rabiei*, Maxwell T. Elling, Alexander J. Herron, Michael Hendrickson

NeurIPS 2024 Workshop ML4PS

Climate PAL: Enhancing Accessibility of Climate Model Data through Conversational AI

Fall 2024

Sonia Crompt, *Behrad Rabiei*, Maxwell T. Elling, Alexander J. Herron, Michael Hendrickson

Meeting of the American Geophysical Union (AGU) 2024

Data PAL: Scientific Data Analysis through Conversational AI

[PDF](#) [↗](#) *Fall 2024*

Sonia Crompt, *Behrad Rabiei*, Maxwell T. Elling, Alexander J. Herron, Michael Hendrickson

Preprint